Changing velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate acceleration and distinguish speeding up, slowing down, and turning.

SUCCESS CRITERIA - I can...

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “acceleration” and “initial” to explain your answer. Sentence starters are optional.

acceleration: rate of change of velocity

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Warm-up: motion and direction

(a) During a shuttle run, an athlete runs *east* at **4 m/s** for **6 s**, then *west* at **3 m/s** for **4 s**. Ignore the brief turn. **Find total distance** and displacement.
(b) A probe's tracking display reports a direction of **30°**. **What else** must the display specify for that angle to be useful?

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: braking on a bike

Reference: $a_{avg} = (v_f - v_i)/\Delta t$.
v_i and v_f are initial and final velocity. Δt is elapsed time. With a steady change in velocity, this gives the constant acceleration.

Worked reasoning: A cyclist brakes while riding *east*. Her speed drops steadily from **8 m/s** to **2 m/s** in **3 s**. Take *east* as positive.
The velocity change is **2** - **8** = **-6 m/s**.
Acceleration = **-6** ÷ **3** = **-2 m/s²**.
She is still moving *east*, but her acceleration points *west*.

Your turn: The **diagram** shows the cyclist’s velocity before and after braking.
1. **Draw an acceleration arrow** below the two velocity arrows.
2. Both velocity arrows point *east*. **Explain why the cyclist is slowing down.**

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Acceleration: add a separate arrow.

Marzano check - circle: ① ② ③

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Continue down each column.

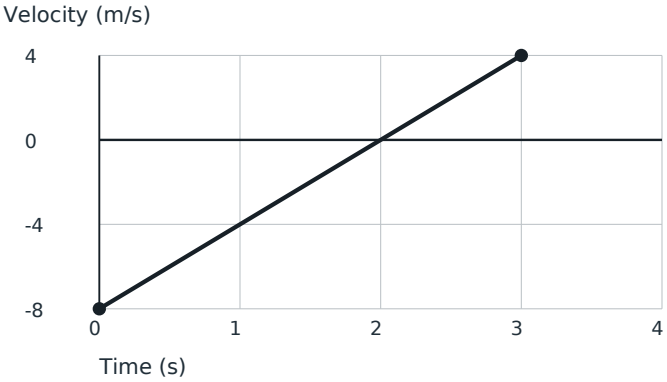
Sentence starters are optional. You may plan in any language.

03 A test car changes direction

The graph shows a remote-control car in a straight-track test. *East* is positive.
(a) **Calculate its acceleration**.
(b) **When is** it momentarily at rest?
(c) **Describe its direction of motion** and whether it speeds up or slows down before and after that instant.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

04 Braking for stopped traffic

A driver sees stopped traffic ahead. The car slows steadily from **24 m/s** to rest in **7.0 s**. Start the clock when braking begins, and take forward as positive.
(a) **Calculate the acceleration**.
(b) **How far** does the car travel while braking?
(c) **Explain why the two answers** have different signs.

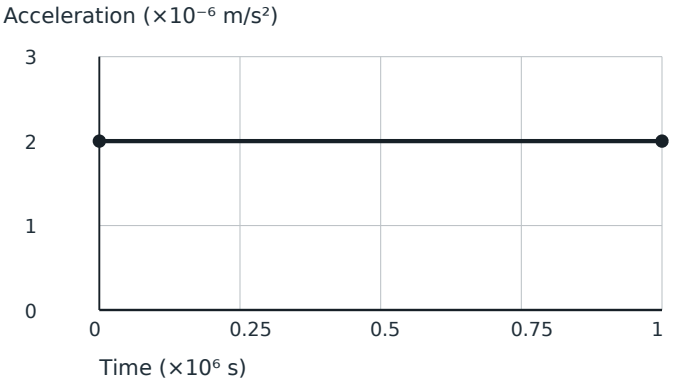
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

05 Can a tiny thruster matter?

A spacecraft's small thruster produces a constant sideways acceleration. Ignore other sideways forces. The graph uses powers of ten on both axes.
(a) **Shade the area under the** line and **calculate the change in sideways velocity**.
(b) Why is the tiny acceleration alone not enough reason to dismiss the thruster?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: 1 2 3

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Continue down each column.

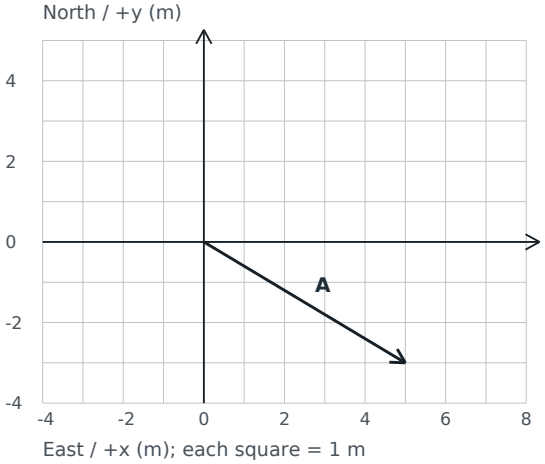
Sentence starters are optional. You may plan in any language.

06 Review: return the robot to its start

A robot's first move is **5 m** *east* and **3 m** *south*, shown as A. Its second move is **3 m** *west* and **7 m** *north*.
(a) **Add the second move head-to-tail** on the grid.
(b) **Find the resultant displacement**: components, magnitude and direction.
(c) **Draw the direct return** to the start and **give its components**.

Word help: resultant = one vector equal to the sum of other vectors

Sentence starter: I found … [unit]. This value means … in this situation.

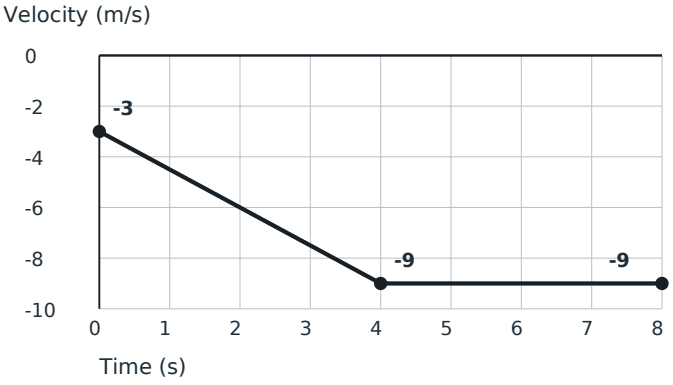


Marzano check - circle: ① ② ③

07 Exit check: thrusters on, then off

The graph shows a simulated spacecraft with its thrusters on from **0** to **4 s**, then off from **4** to **8 s**. *East* is positive. For each interval, calculate acceleration and **describe its direction of motion** and whether it speeds up, slows down or keeps a steady speed.

Sentence starter: I found … [unit]. This value means … in this situation.

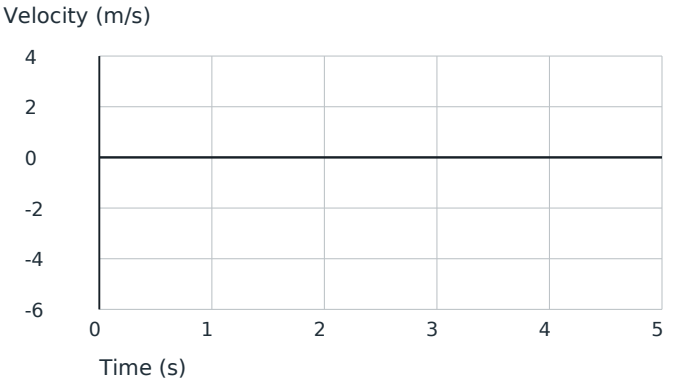


Marzano check - circle: ① ② ③

08 STRETCH: past the turning point

A remote-control car initially travels *west* at **6 m/s**. It then accelerates at **2 m/s²** *east* for **5 s**, including after it reverses. *East* is positive.
(a) **Complete the graph** and mark when it reverses.
(b) Use the signed areas to **find displacement** and total distance. **Explain why they differ**.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③